Lubrificantes

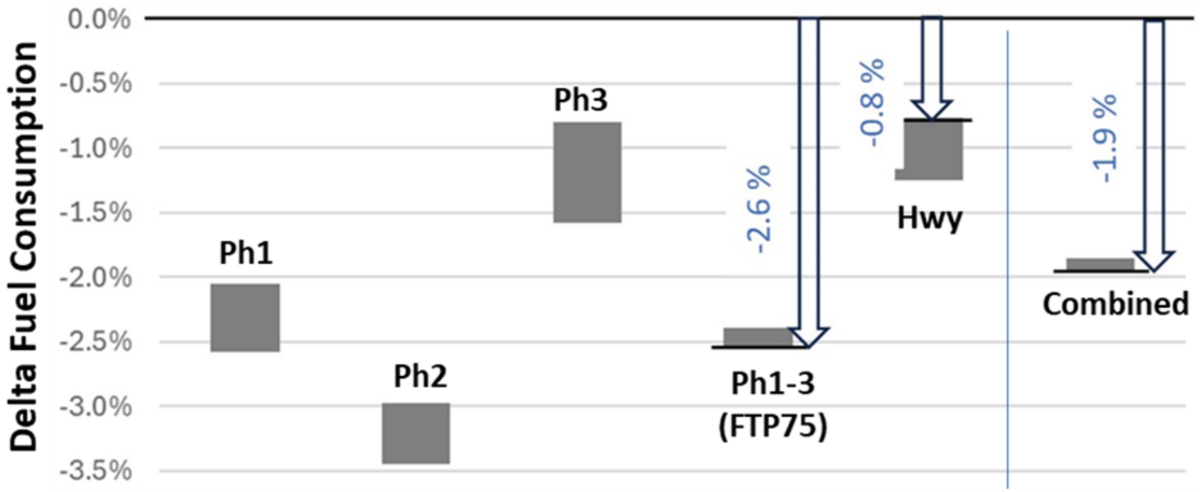
dezembro de 2024

P&D

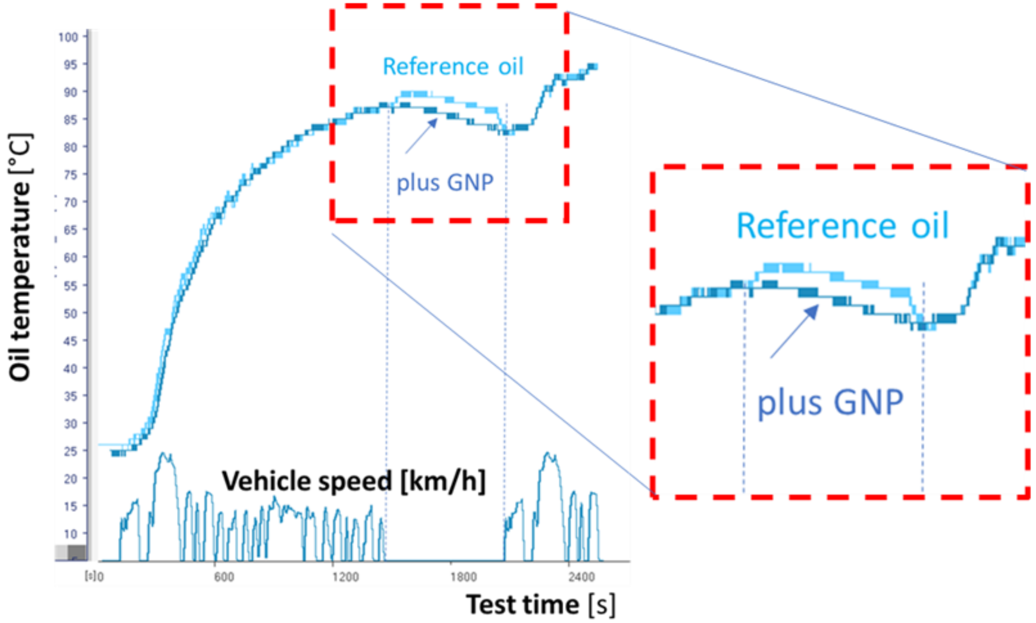
GERDAU GRAPHENE

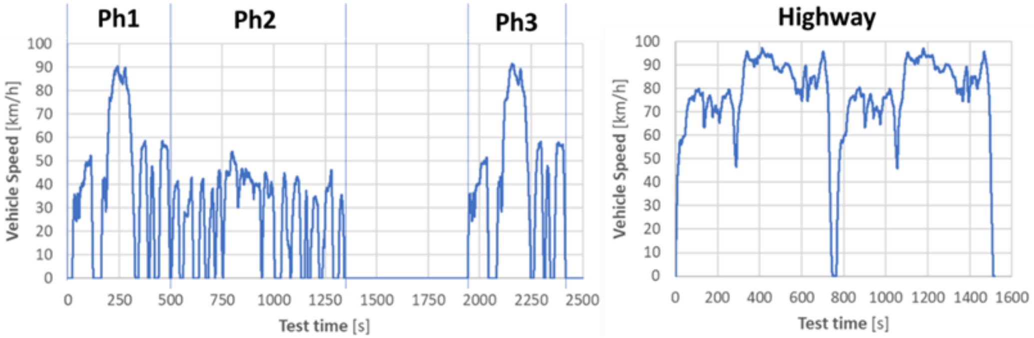
Tests at VW (with L66_2, 0.1%)

2.6% of fuel saving at Urban conditions, 0.8% at highway



Faster cooling. Higher thermal conductivity?





	Units	Ph1 & 3	Ph2	Highway
Duration	s	505	864	12.75
Distance	km	5.78	6.21	16.58
Mean Velocity	km/h	41.20	25.88	77.73
Max. Velocity	km/h	91.2	55.2	96.56
Stops		6	13	None

Paper submitted with VW and Idemitsu, and USP

eng

MDPI

Article

Effect of Graphene nanoplatelets as lubricant additive on fuel consumption during vehicle emission tests.

Eduardo Tomanik[1*], Wania Christinelli[1], Pamela Sierra Garcia[1], Scott Rajala[2], Jesuel Crepaldi[3], Davi Franzosi[4], Roberto Martins Souza[4], Fernando Fusco Rovai[5,6]

[1] Gerdau Graphene Brazil; eduardo.tomanik@ex.gerdaugraphene.com, wania.christinelli@gerdaugraphene.compamela.garcia@gerdaugraphene.com
[2] Idemitsu Lubricants America Corp.; srajala.1460@idemitsu.com
[3] Idemitsu Lube South America Ltda; jesuel.crepaldi.0450@idemitsu.com
[4] University of São Paulo; robertosouza@usp.br, davi.franzosi.oliveira@usp.br
[5] VW do Brasil – Way to Zero Center; fernando.rovai@volkswagen.com.br
[6] Department of Mechanical Engineering, Centro Universitário FEI; ffusco@fei.edu.br
* Correspondence: Eduardo.tomanik@gmail.com

Abstract: Lubricant friction modifier additives are used on lower-viscosity engine oils to mitigate the impact on the boundary friction. In this work, the use of Graphene nanoplatelets, with an average of 9 layers, was first investigated in a fully formulated SAE 0W-20 engine oil through the use of reciprocating tribological tests at 40 and 120 °C and Hertzian pressure up to 1.2 GPa. Then, engine fuel consumption was evaluated through fuel consumption in a vehicle emission test adding 0.1% of GNP, in an improved booster, on a fully formulated 5W-20 SAE oil. Adding 0.1% of GNP reduced fuel consumption by 2.6% in the FTP-75 cycle and 0.8% in the Highway.

CM018 – Desenvolvimento de aditivo químico base água contendo grafeno para ativação da escória em baixas idades para ganhos de resistência

Atualizações do desenvolvimento

dezembro de 2024

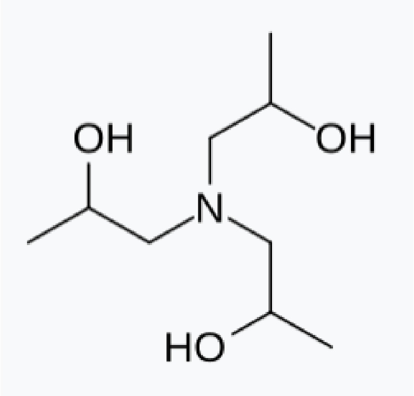
P&D

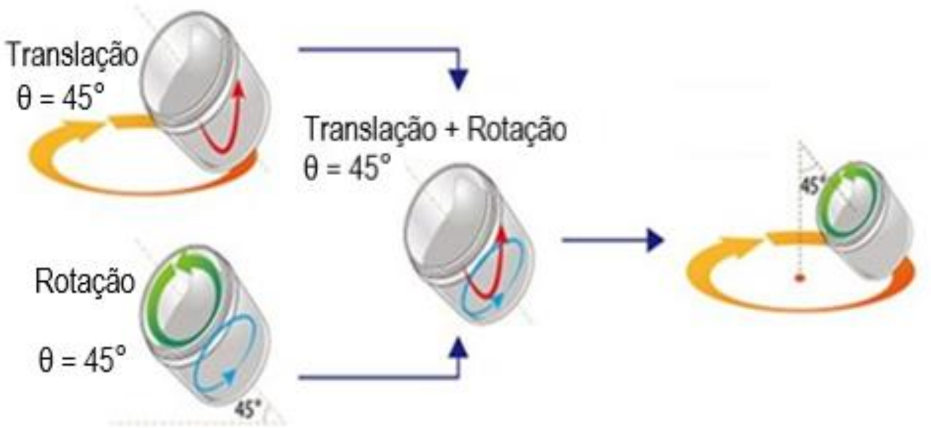
GERDAU GRAPHENE

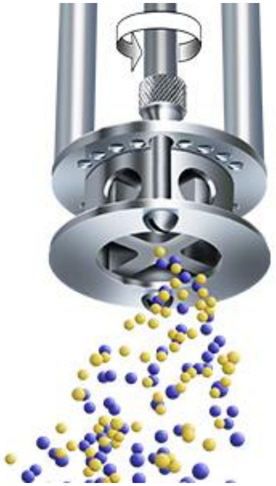
INFORMAÇÕES DO PROJETO	
Código & Nome do Projeto	CM018 – Aditivo Químico base água contendo Gr para ativação da escória em baixas idades para ganhos de resistência
Data de início de apropriação do ativo intangível	mai/24
Data do documento	mai/24
Plataforma	Construção
Segmento de Mercado	Construção Civil
Aplicações	Concretos estruturais com uso de adição mineral à base de escória de alto forno

PROCESSO DE PRODUÇÃO

+





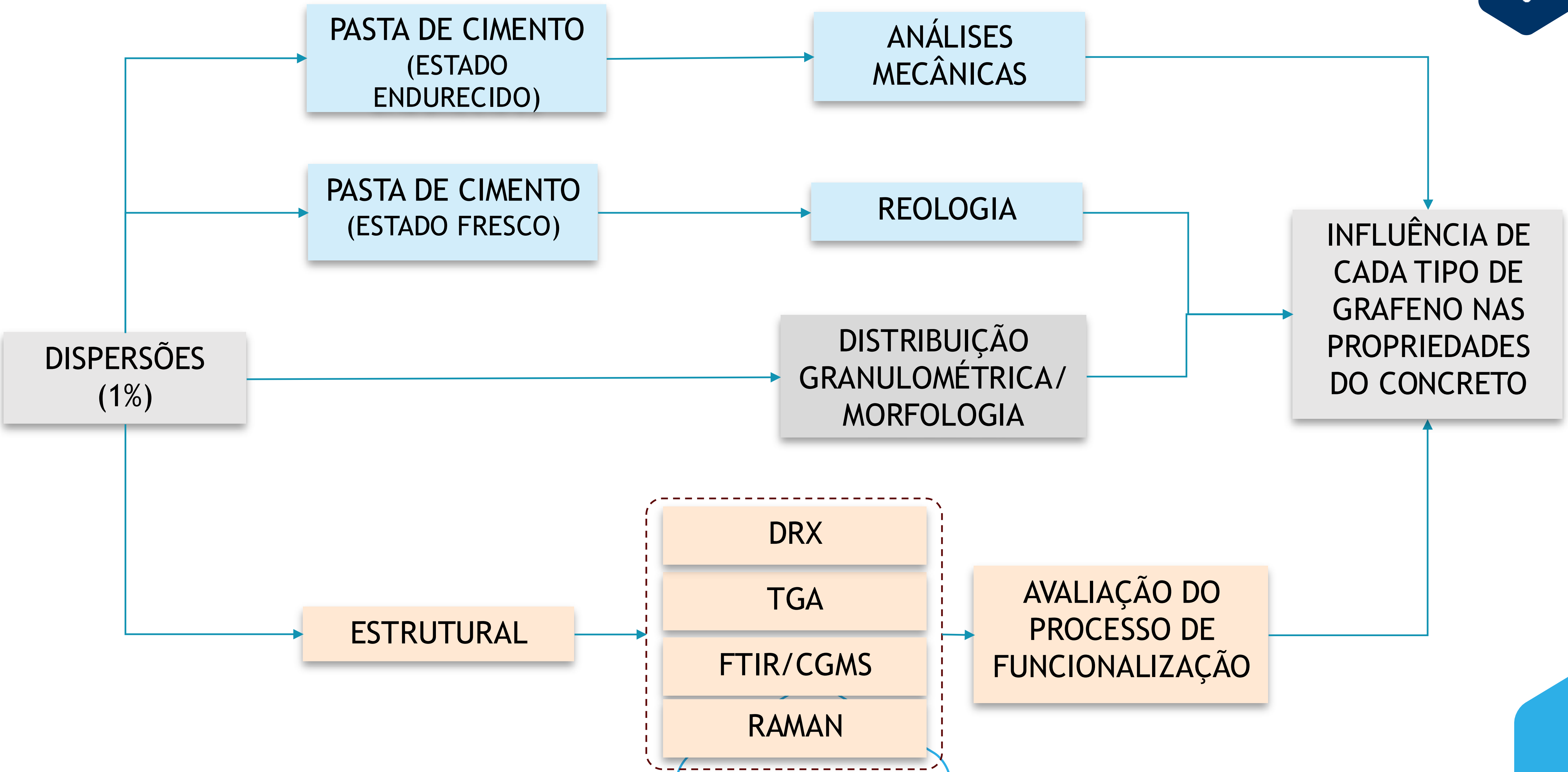


GRAFENOS AVALIADOS

0X	-	NanoXplore
BS3	-	BlackSwan
BS8	-	BlackSwan
FG05	-	First Graphene
FG10	-	First Graphene
FG20	-	First Graphene
HP10	-	Hexo
HP50	-	Hexo

A escolha inicial destes levou em conta a disponibilidade e morfologia

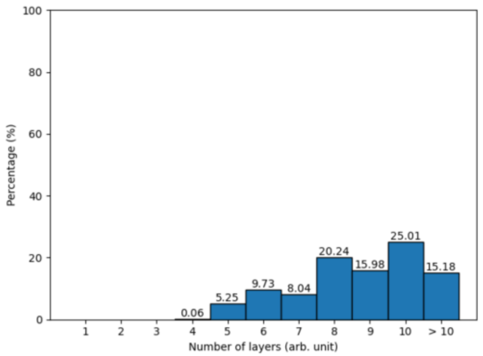
DESENVOLVIMENTO DE CONCENTRADOS DE GRAFENO

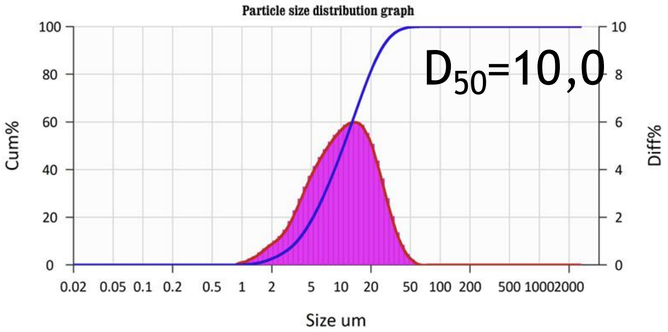


Concentrados

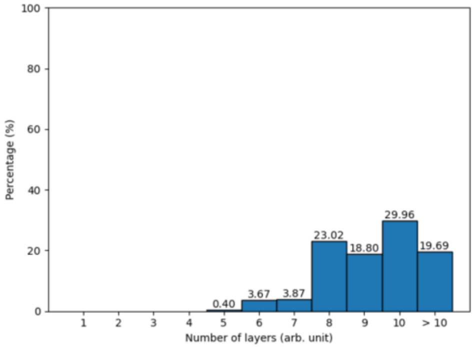
CM018

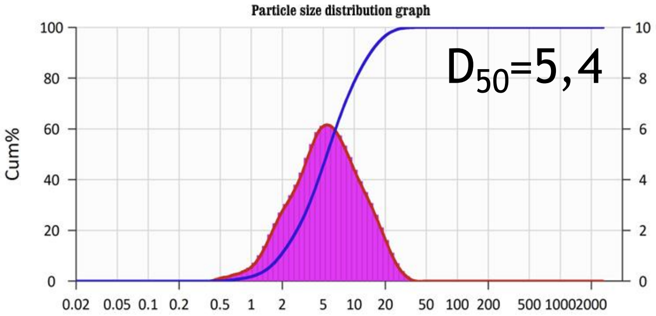
FG10



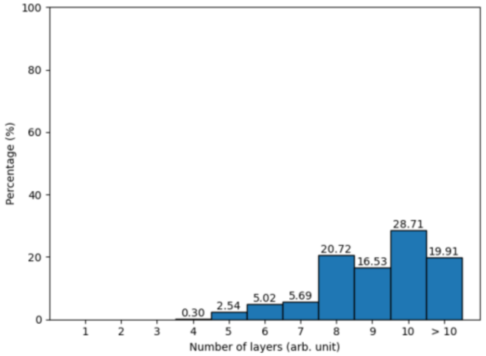


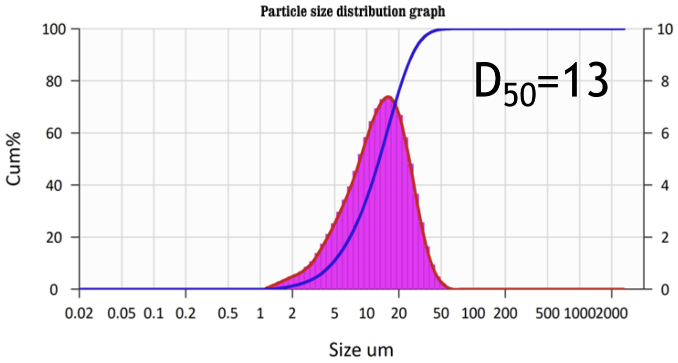
BS3



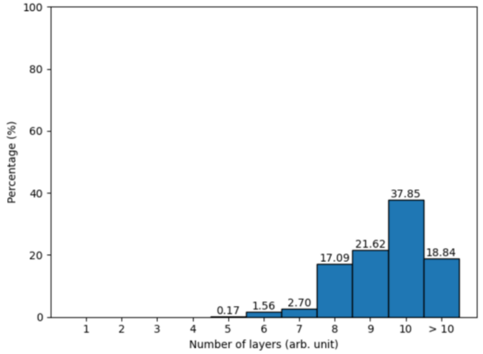


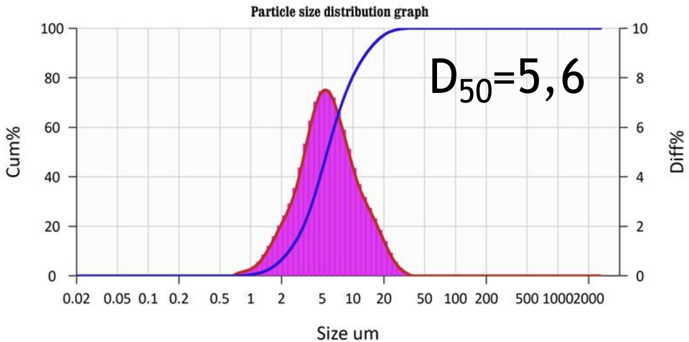
0X





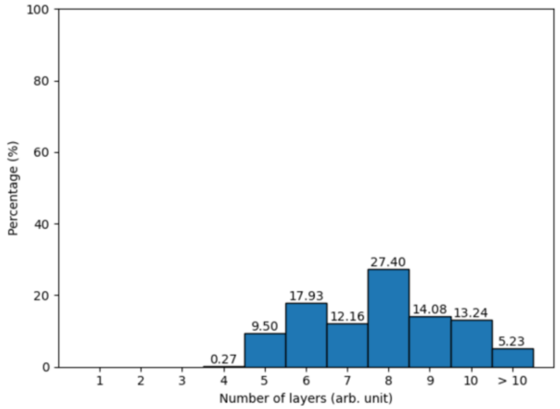
HP10

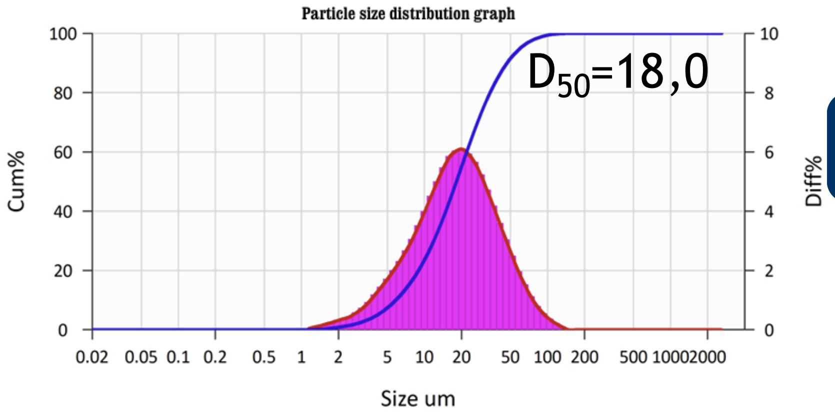




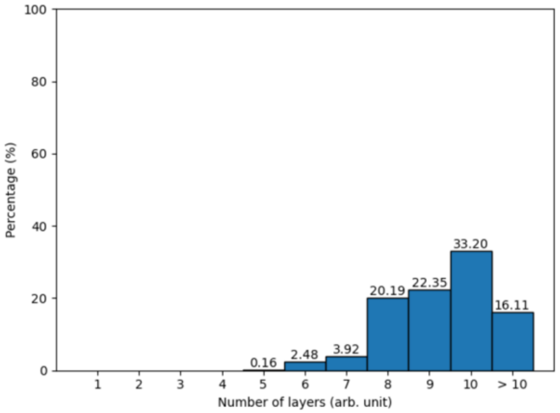
CM018

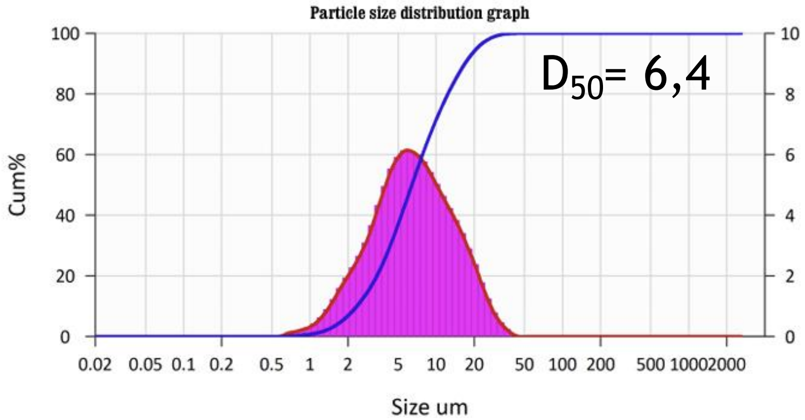
FG20



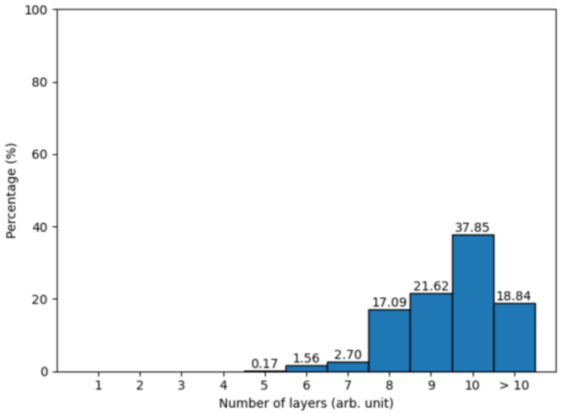


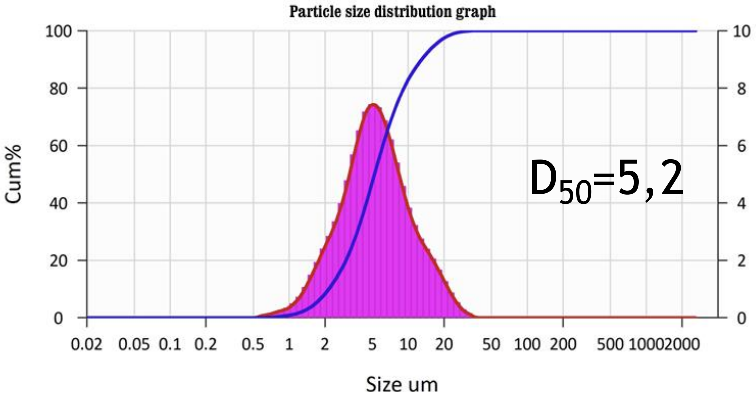
BS8





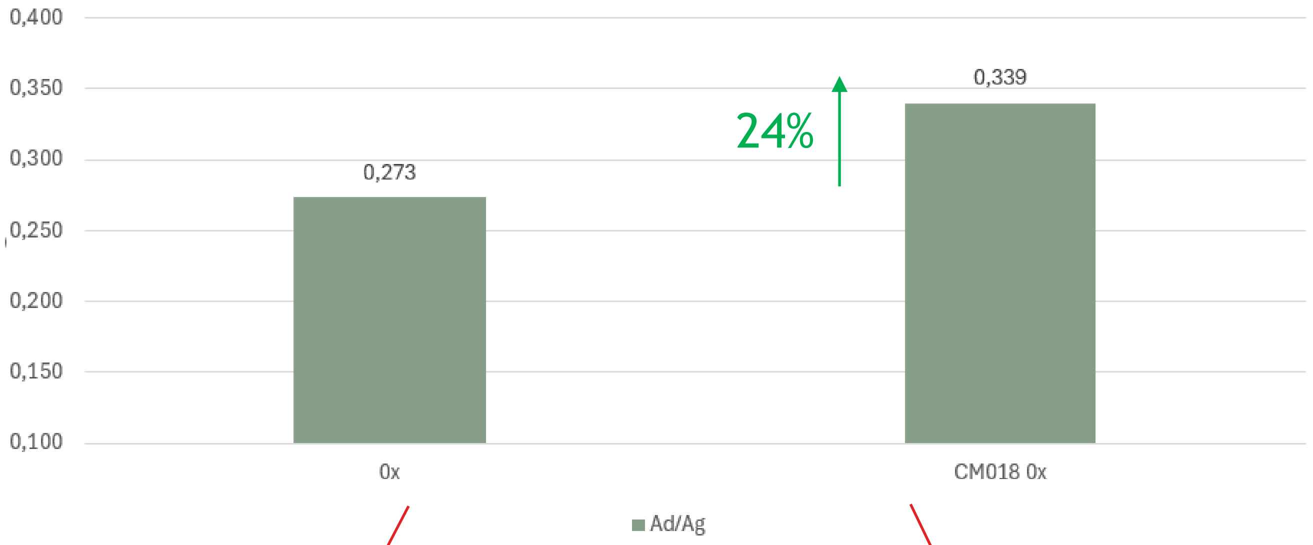
HP50

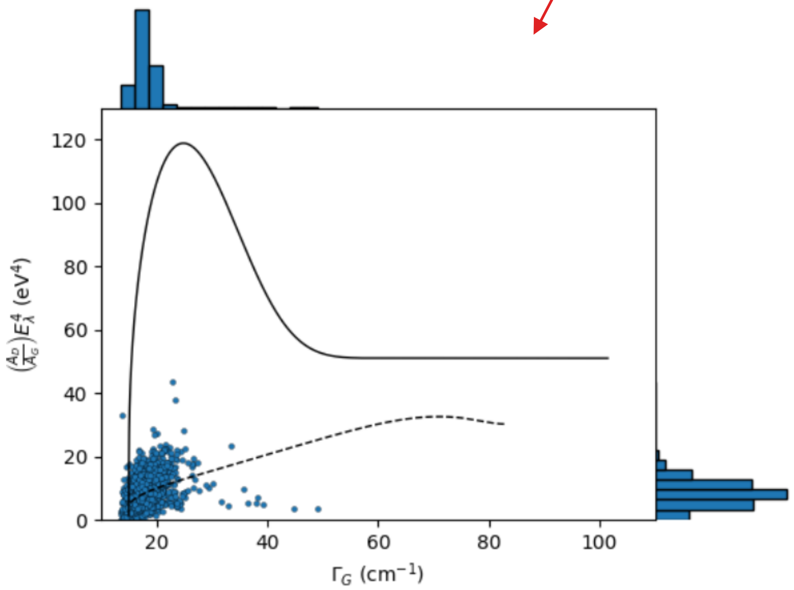


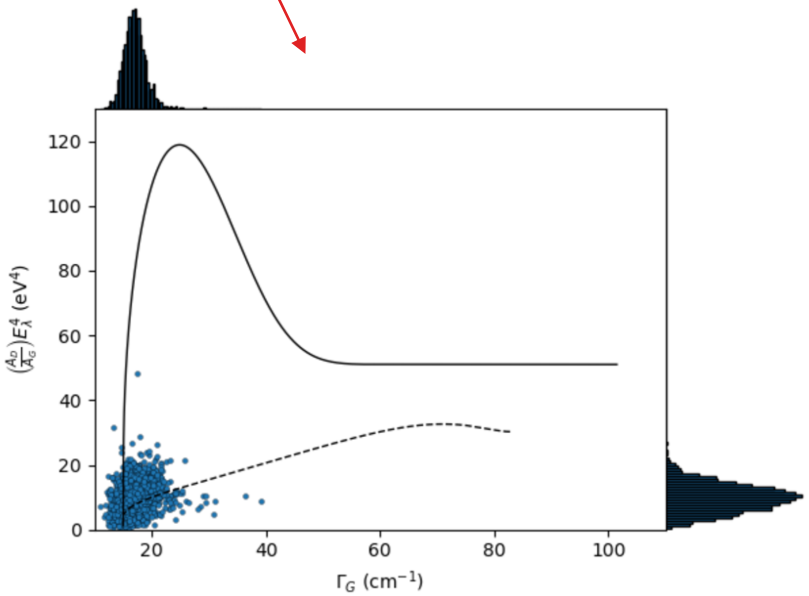


RAMAN

A_D/A_G

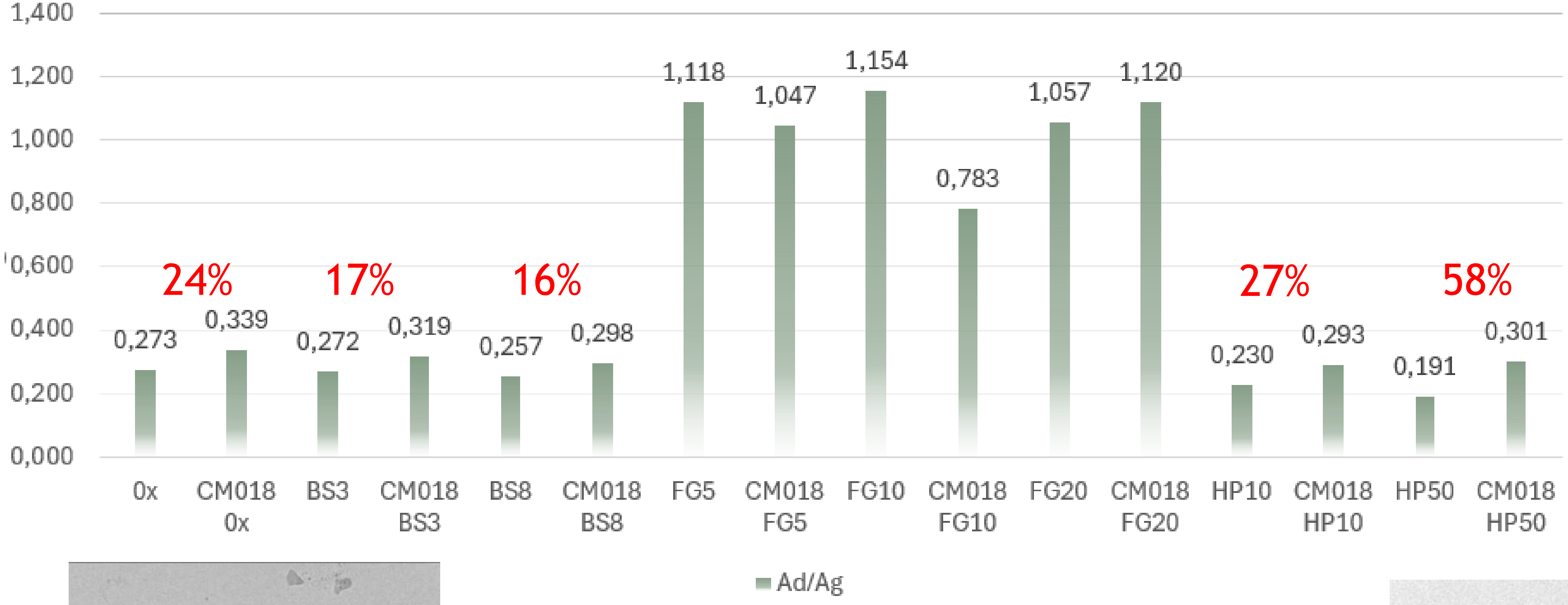

0,400
0,350
0,300
0,250
0,200
0,150
0,100
0,273
0,339
24%
0x
CM018 0x
Ad/Ag

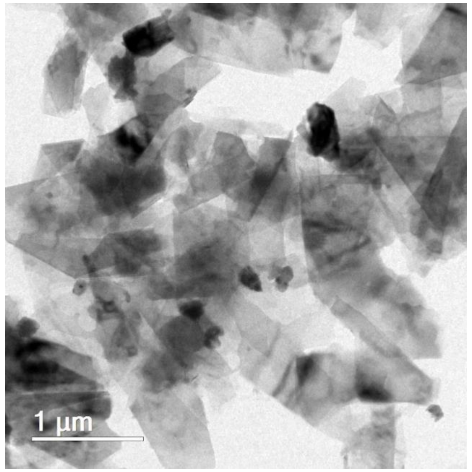

$\left(\frac{A_D}{A_G}\right)E_\lambda^4$ (eV^4)
Γ_G (cm^{-1})


$\left(\frac{A_D}{A_G}\right)E_\lambda^4$ (eV^4)
Γ_G (cm^{-1})

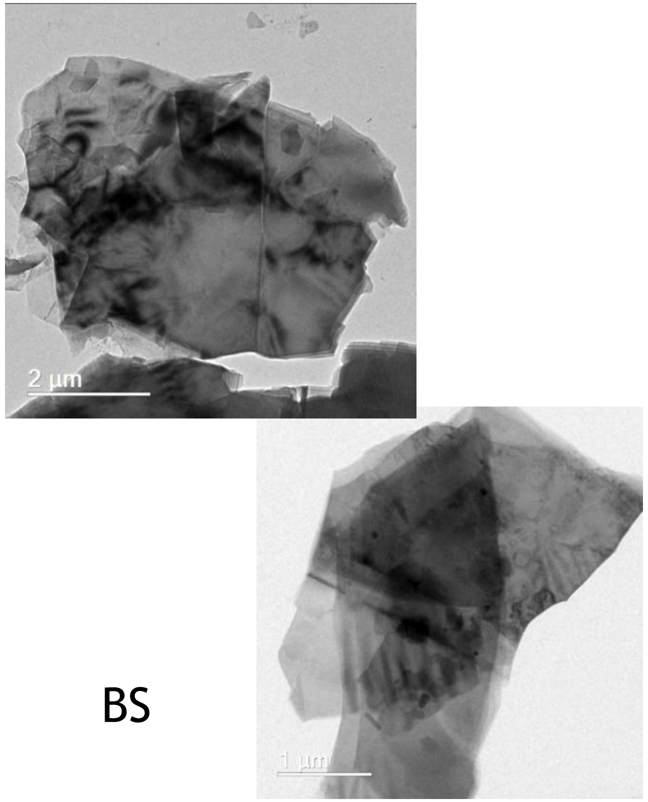
RAMAN

A_D/A_G

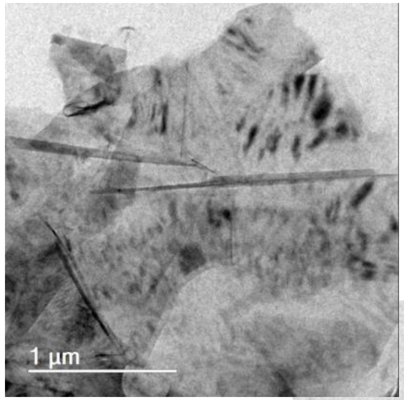


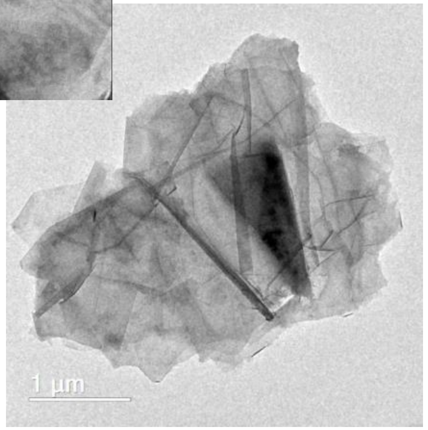


0x

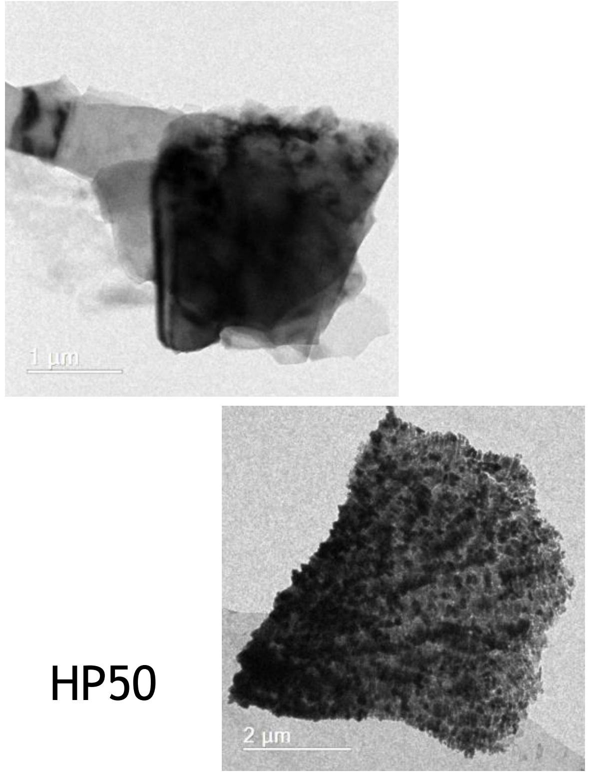


BS



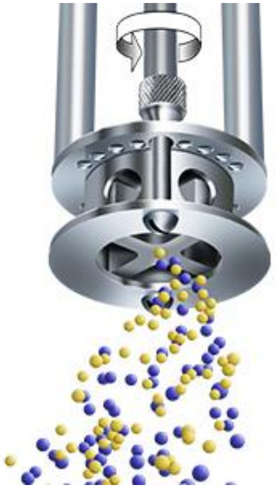


FG

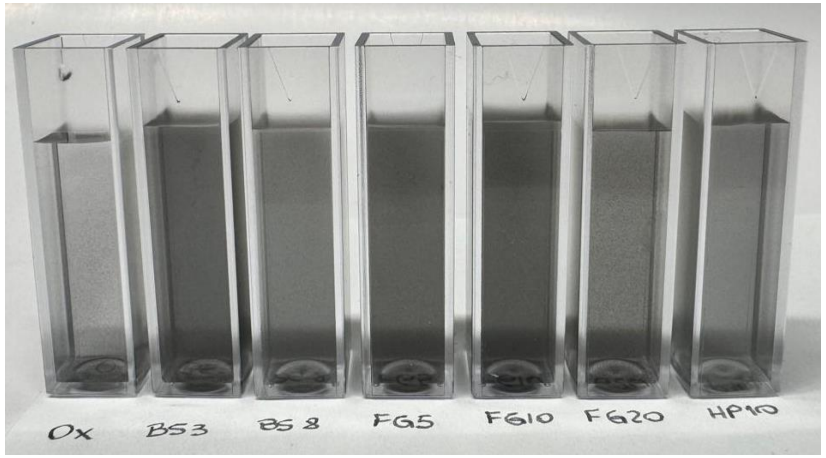


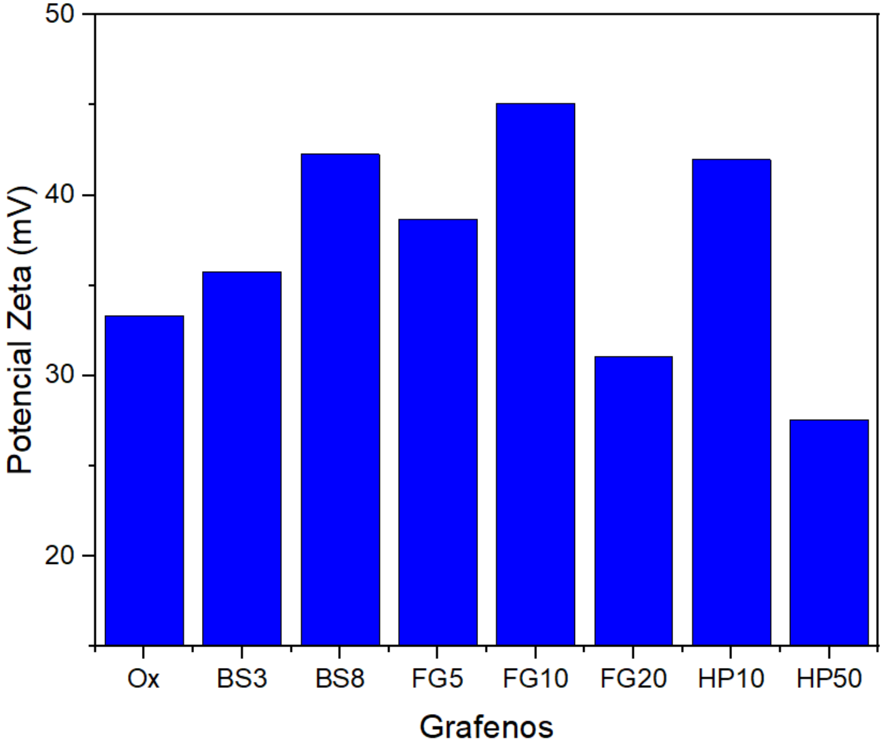
HP50

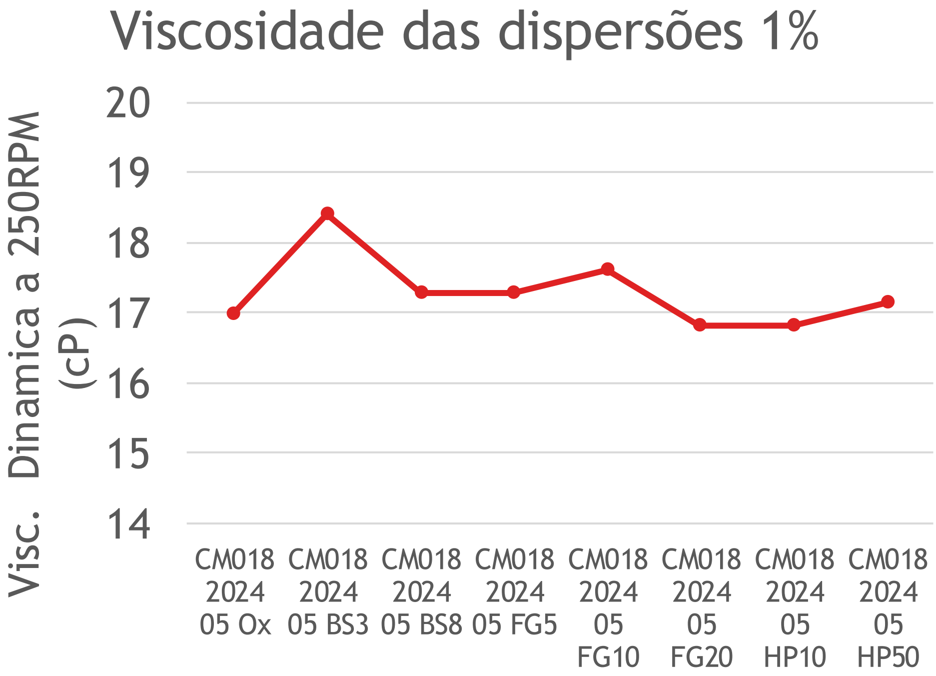
DISPERSÕES 1%



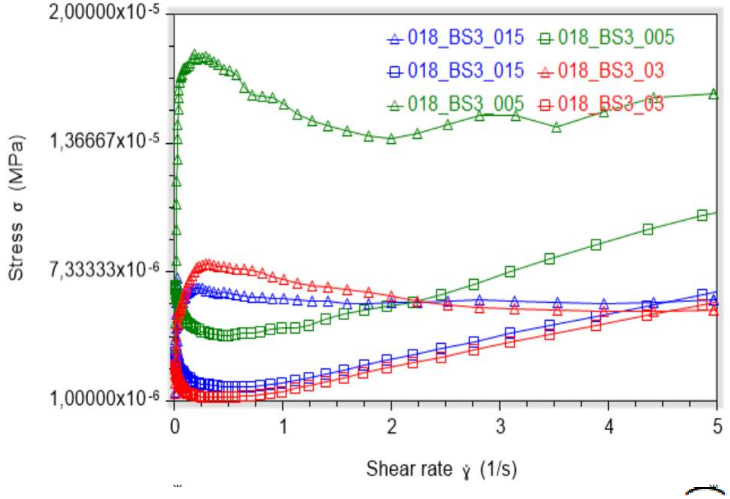
3000 RPM
5min

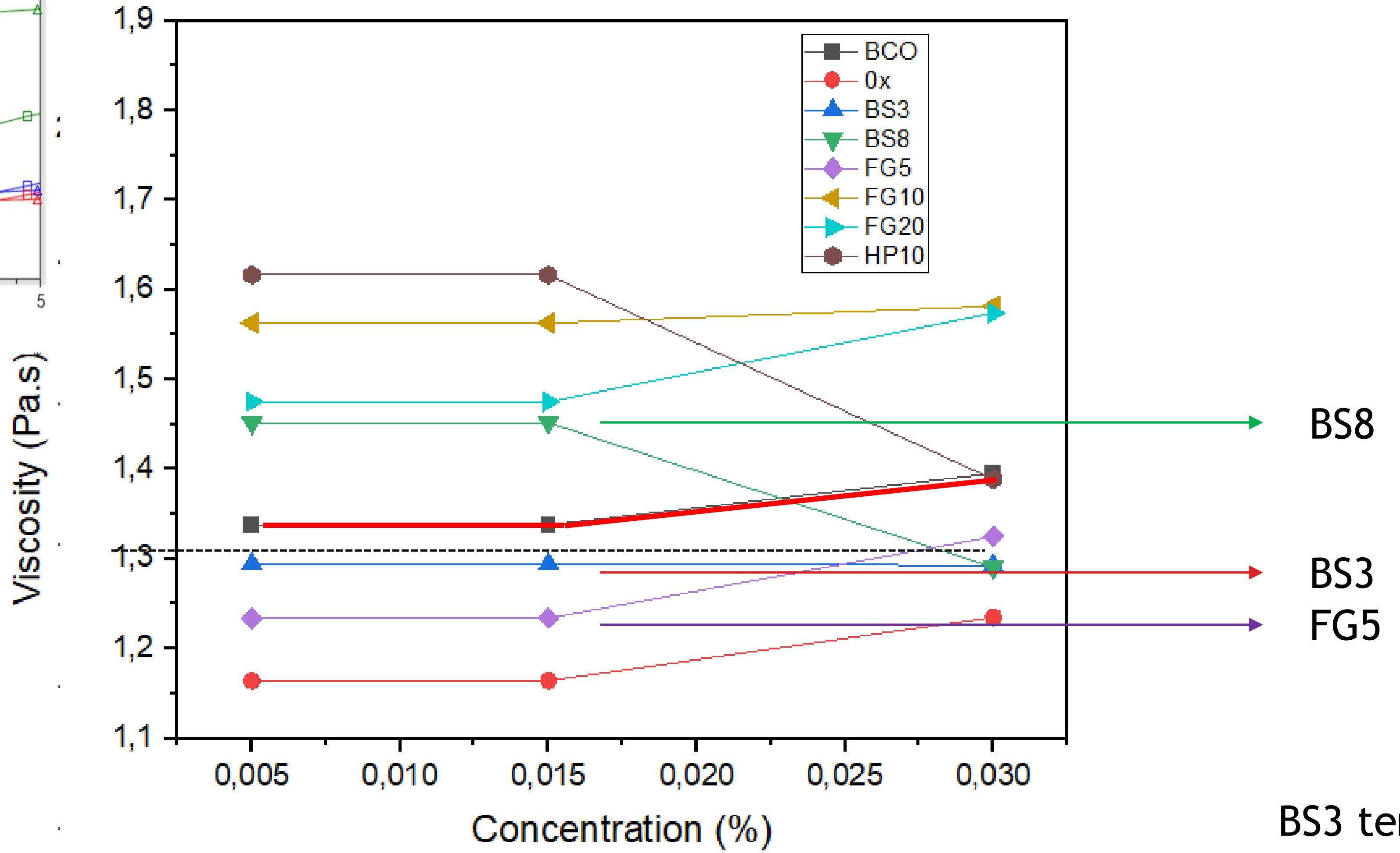






CARACTERIZAÇÕES
Flow curve





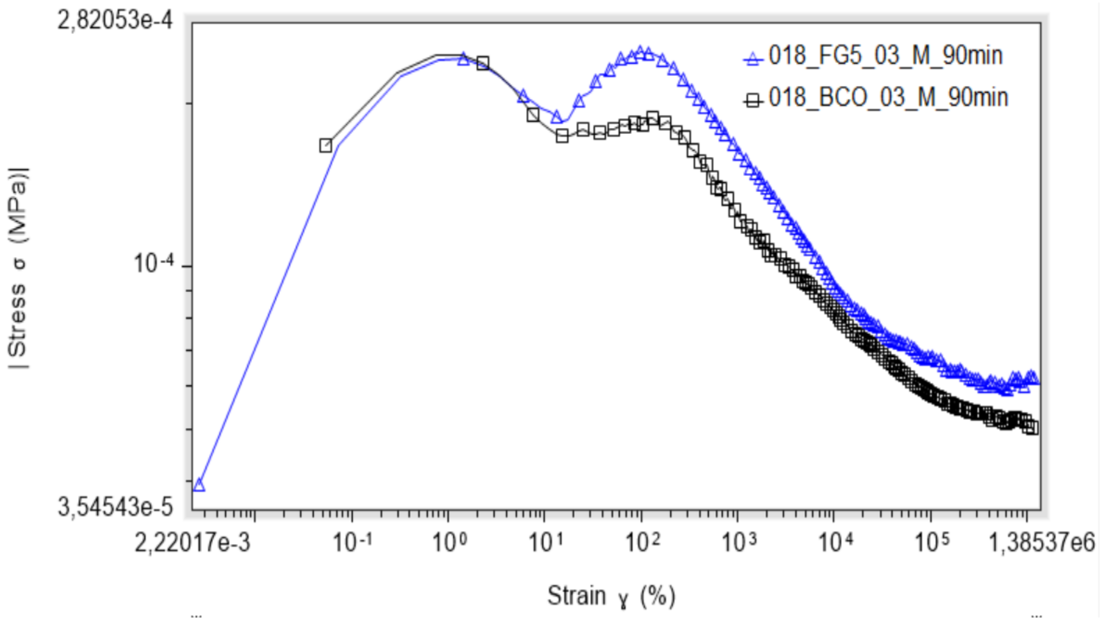
BS3 tem menor influência na reologia no estado fresco

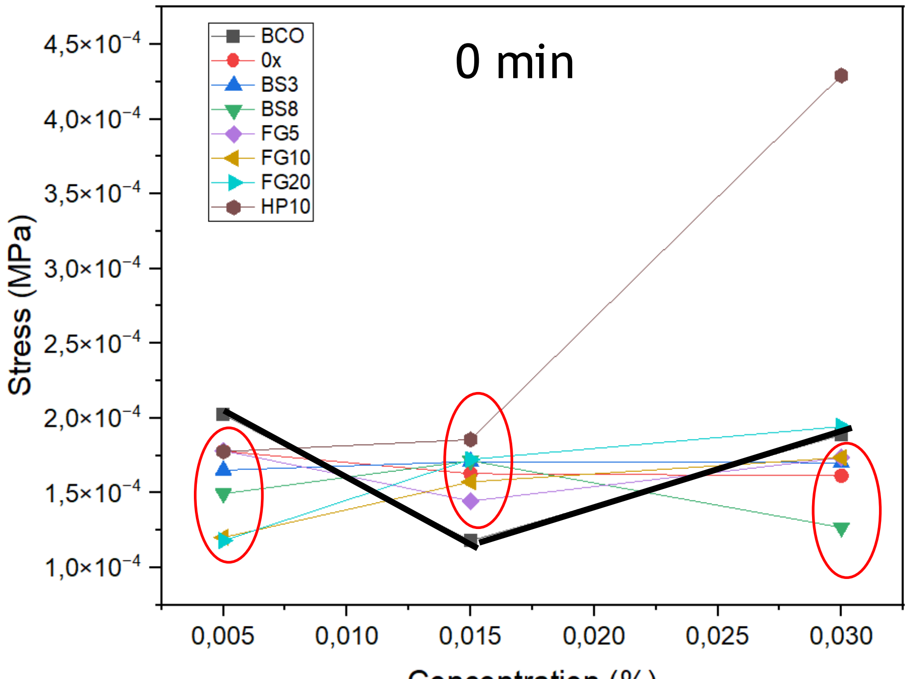
Reologia em pasta de cimento

CARACTERIZAÇÕES

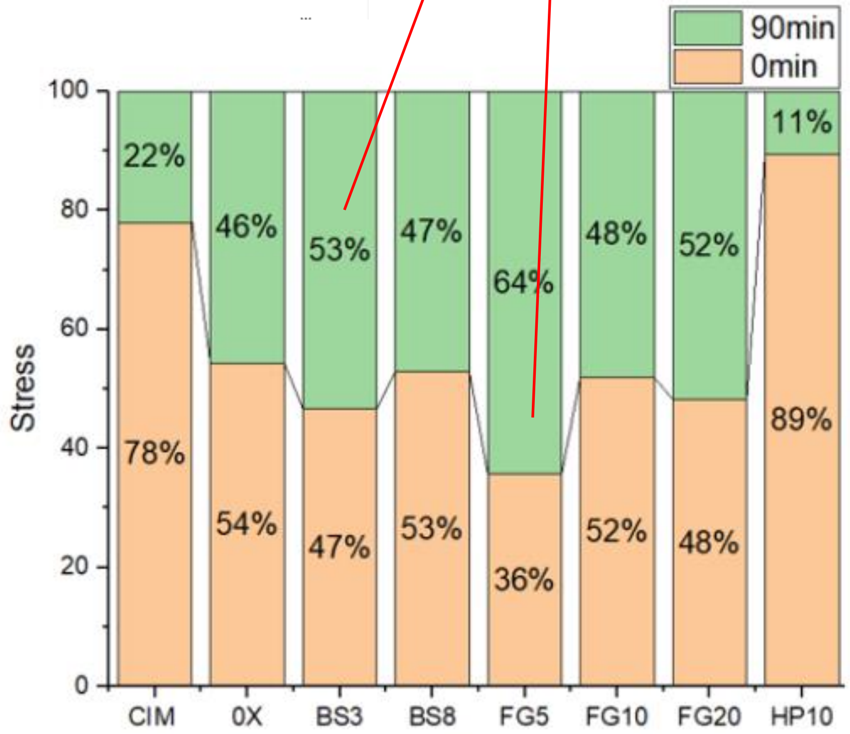
Stress Growth

Pasta de cimento





BS3 FG5



0,03%

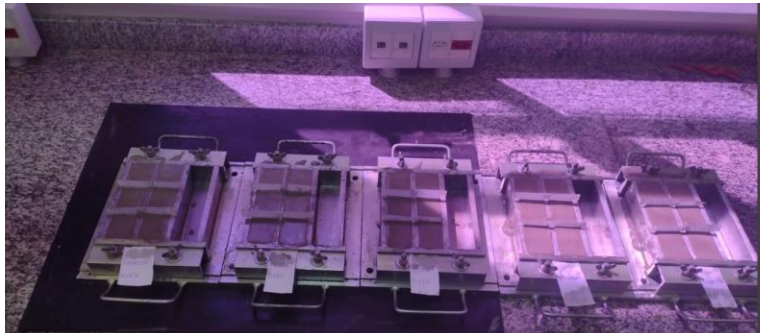
This content is Internal.

AVALIAÇÃO DE DESEMPENHO

Realizada em corpos de prova de pasta de cimento, sendo estes produzidos na forma prismática e com dimensões de 40 x 40 x 40 mm.

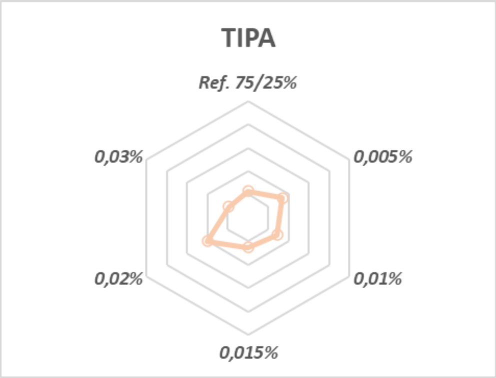
O programa de testes contou com a produção das pastas de cimento com teores de Grafeno de:

- 0,005%
- 0,010%
- 0,015%
- 0,020%
- 0,030%

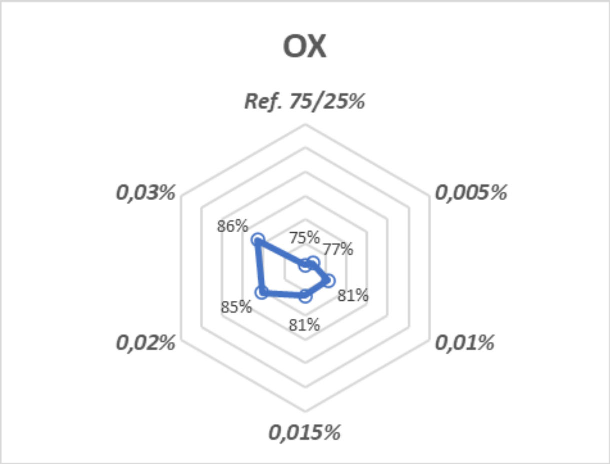


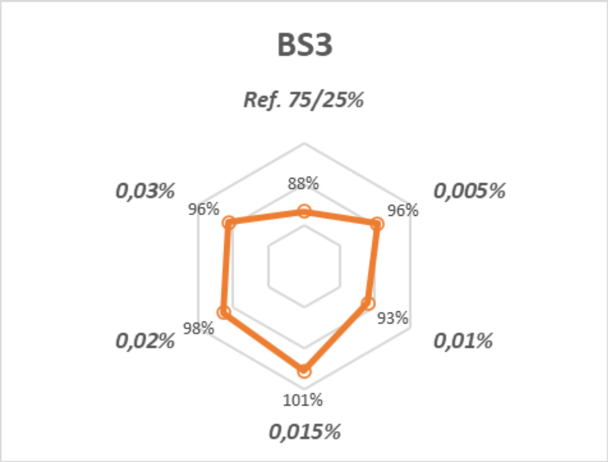
AVALIAÇÃO DE DESEMPENHO

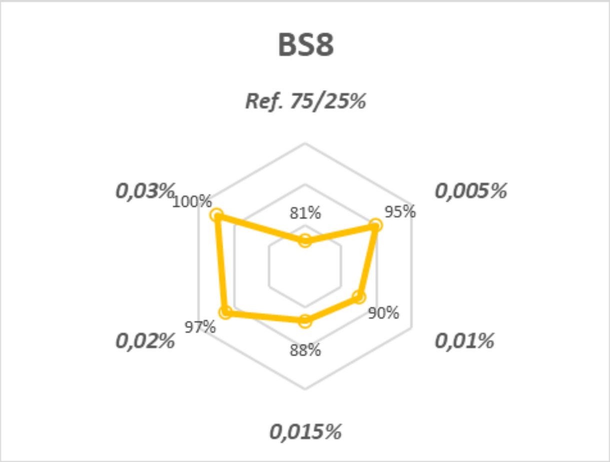
Resultados – 3 dias

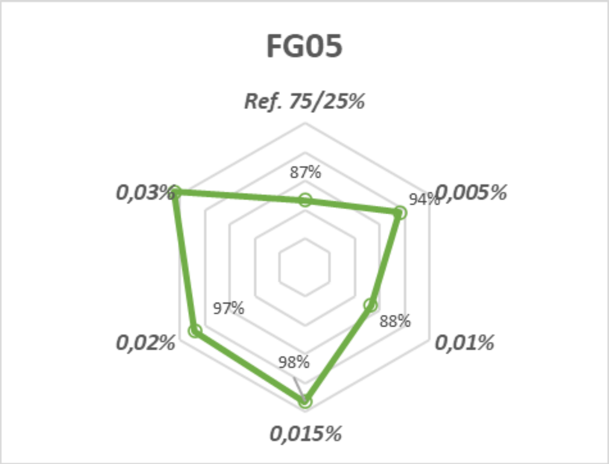


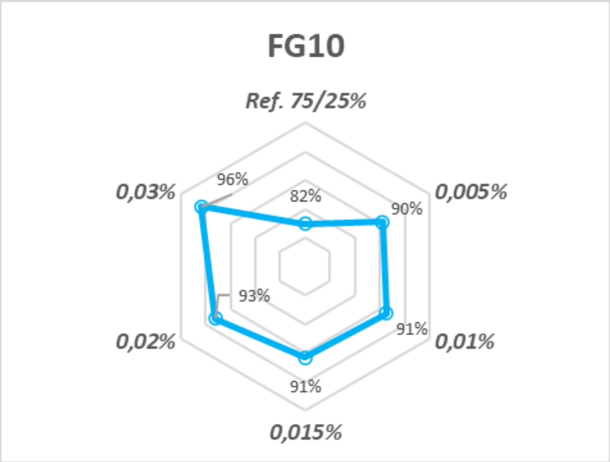
Percentual de resistência à compressão em relação a referência (100% cimento)

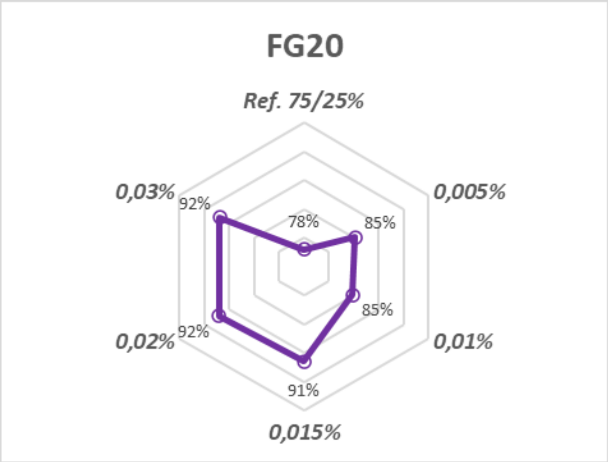


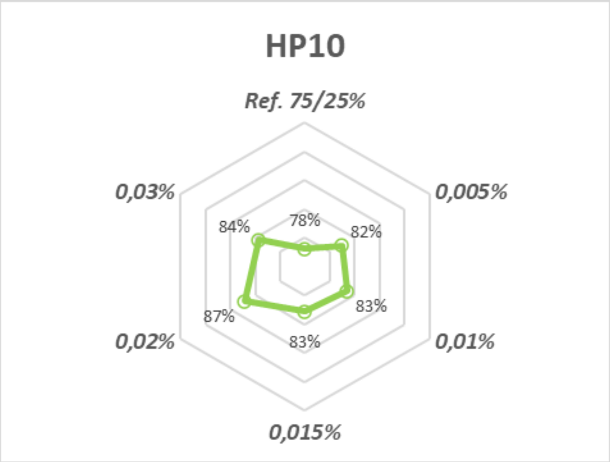


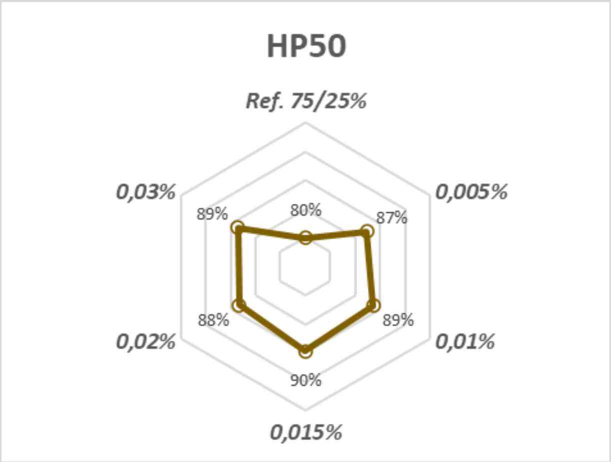






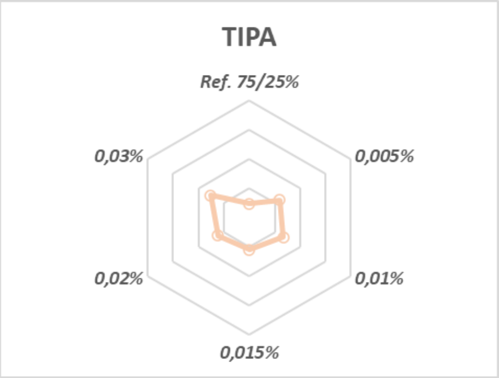




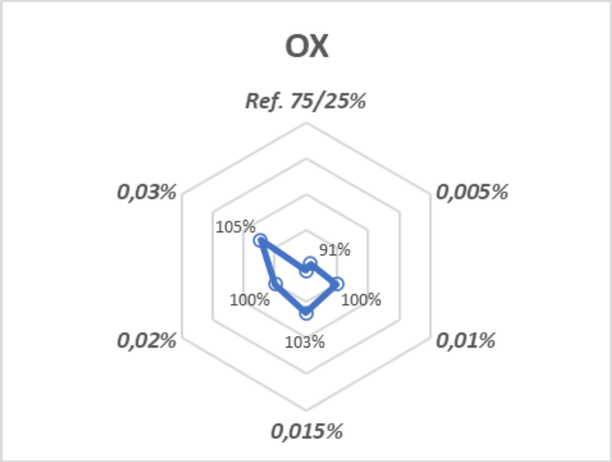


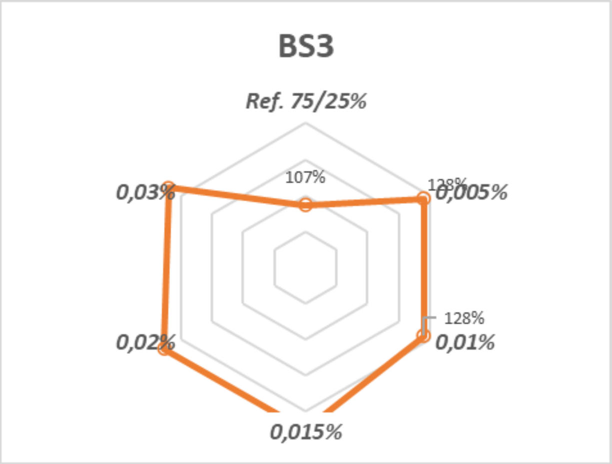
AVALIAÇÃO DE DESEMPENHO

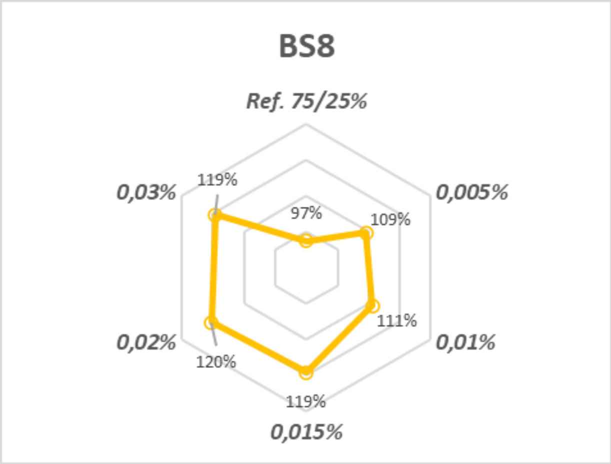
Resultados – 7 dias

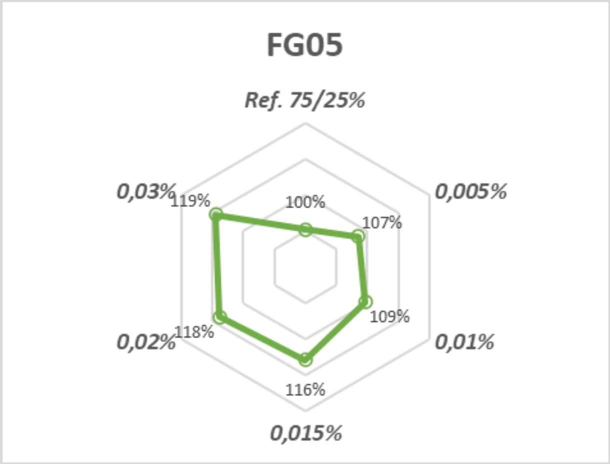


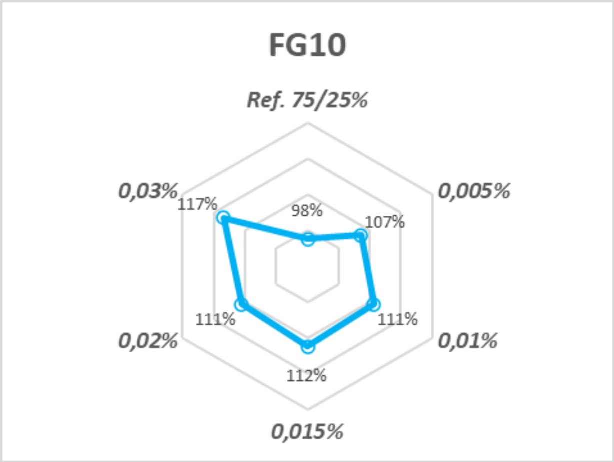
Percentual de resistência à compressão em relação a referência (100% cimento)

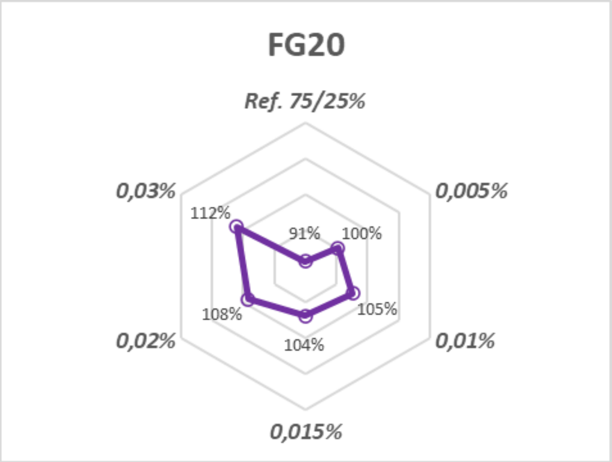


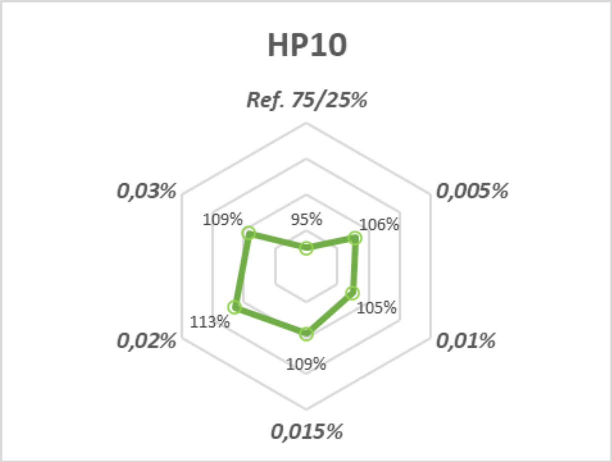


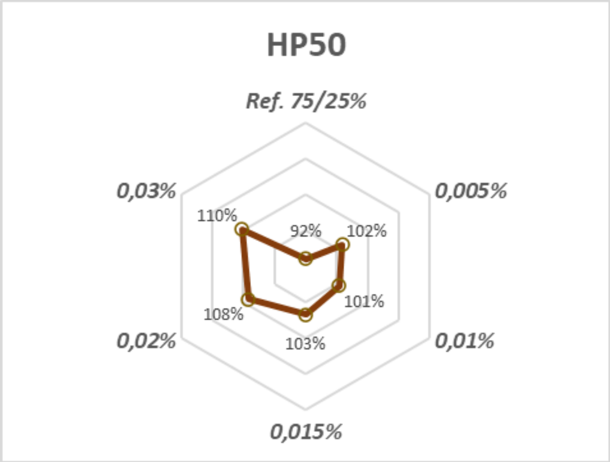








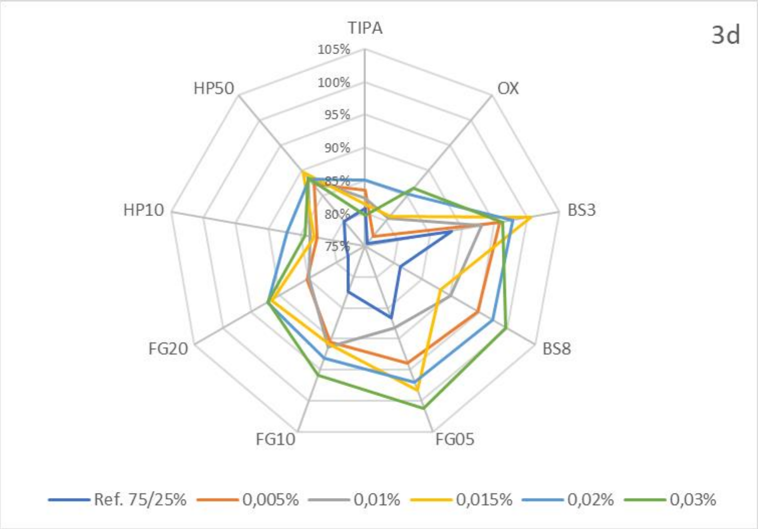


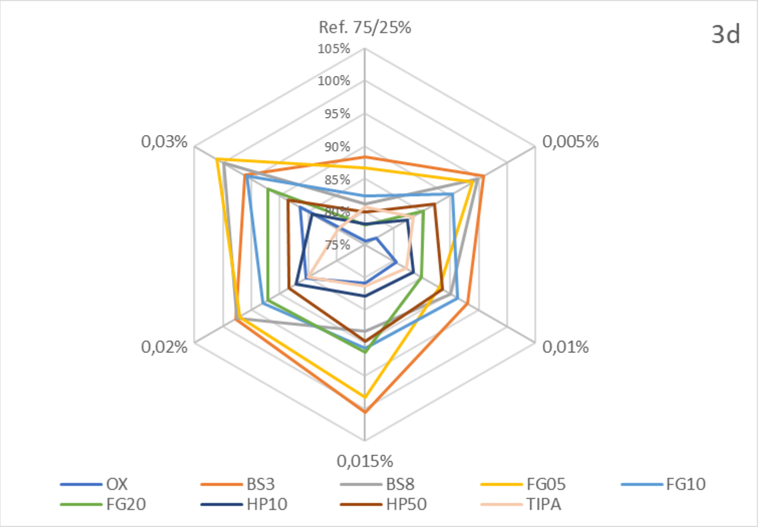


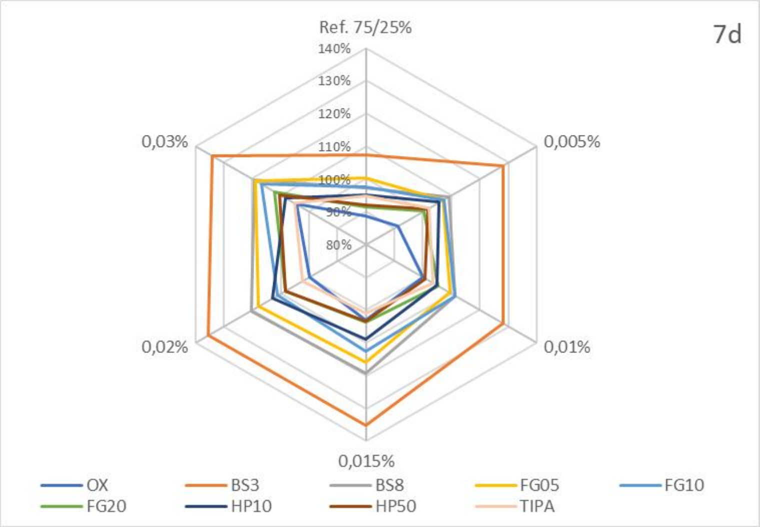
AVALIAÇÃO DE DESEMPENHO

Resultados

Percentual de resistência à compressão em relação a referência (100% cimento)







GERDAU
GRAPHENE

Obrigado!